

Week 4: The six-vertex model I

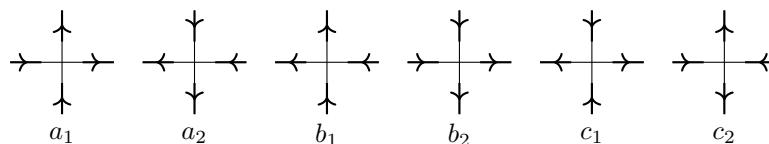
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1 Introduction

On a subset of the square lattice, label each edge with an arrow such that at any vertex, the number of incoming arrows is equal to the number of outgoing arrows.

There are six possible configurations, which each get a weight:



In the presence of an electric field with E_x denoting the horizontal field strength and E_y denoting the vertical field strength, these weights are

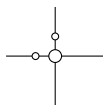
$$\begin{aligned} a_1 &= ae^{E_x + E_y}, & a_2 &= ae^{-E_x - E_y}, \\ b_1 &= be^{-E_x + E_y}, & b_2 &= be^{E_x - E_y}, \\ c_1 &= ce^\gamma, & c_2 &= ce^{-\gamma}. \end{aligned}$$

Here γ is an additional parameter for the two c -weights. We will make the zero field assumption: $a_1 = a_2 = a$, $b_1 = b_2 = b$, $c_1 = c_2 = c$.

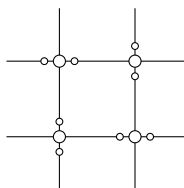
For example,

$$w \left(\begin{array}{cc} \begin{array}{c} \xrightarrow{a_1} \bullet \xrightarrow{b_2} \\ \xleftarrow{c_2} \bullet \xleftarrow{c_1} \end{array} & \begin{array}{c} \xrightarrow{b_2} \bullet \xrightarrow{a_1} \\ \xleftarrow{c_1} \bullet \xleftarrow{c_2} \end{array} \\ \hline \begin{array}{c} \xrightarrow{c_2} \bullet \xrightarrow{c_1} \\ \xleftarrow{a_1} \bullet \xleftarrow{b_2} \end{array} & \begin{array}{c} \xrightarrow{a_1} \bullet \xrightarrow{b_2} \\ \xleftarrow{b_2} \bullet \xleftarrow{a_1} \end{array} \end{array} \right) = a_1 b_2 c_1 c_2.$$

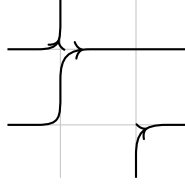
There are other equivalent versions of the six-vertex model. One is called square ice: it puts a “molecule” at each vertex, where a molecule looks like:



There is one hydrogen on each edge, close to one of its endpoints, and exactly two hydrogens close to each oxygen. If two molecules are adjacent, then their hydrogens are not allowed to point toward each other. An example of a square ice model is:



There is also a path model, which models a collection of osculating paths on a square grid. Keep just the edges whose arrows point right or up. These form north-east going paths which can meet at a vertex, but cannot share an edge. At a vertex with four occupied edges, pair the west edge with the north edge and the south edge with the east edge. Here is an example:



Definition 1.1. The **partition function** is

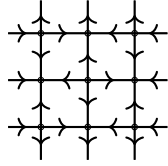
$$Z = \sum_{\text{configurations}} \prod_v w(v)$$

where $w(v)$ denotes the weight of a vertex v .

There are many interpretations of Z as counting combinatorial objects. Here are three examples, with domain-wall boundary conditions (horizontal boundary arrows pointing in and vertical ones pointing out):

0	1	2	0
1	2	1	2
2	1	2	1
0	2	1	0

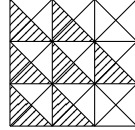
$$3Z = \#\{3\text{-colorings}\}$$



$\longleftrightarrow$

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & -1 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$Z = \#\{\text{ASMs}\}$$



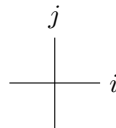
$$Z = \#\{\text{domino tilings}\}$$

For the first two examples take $a = b = c = 1$. The colorings are of faces, with colors in $\mathbb{Z}/3\mathbb{Z}$; the colors on opposite sides of an arrow differ by 1, and choosing one face color gives the factor of 3. For domino tilings of the Aztec diamond of order $n - 1$, take an $n \times n$ grid with $a = b = 1$, $c_1 = 2$, $c_2 = 1$: one of the c -vertices has two resolutions. Equivalently, at zero field take $a = b = 1$, $c = \sqrt{2}$, with an overall boundary-dependent factor.

2 Transfer-matrix formalism

We can recast Z with a transfer matrix. Assume we have the topology of a torus, with m rows and ℓ columns. Then Z will be the trace of a matrix T_ℓ^m .

To define it, at a crossing of lines i and j



we will define a local transfer matrix. Attach a vector space V_i to the line i and V_j to line j .

$$V_i = \text{span}\{\rightarrow, \leftarrow\}$$

$$V_j = \text{span}\{\uparrow, \downarrow\}.$$

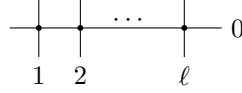
Then R_{ij} is a linear map from $V_i \otimes V_j \rightarrow V_i \otimes V_j$. We think of $V_i \otimes V_j$ as spanned by

$$\rightarrow\uparrow, \quad \rightarrow\downarrow, \quad \leftarrow\uparrow, \quad \leftarrow\downarrow.$$

The left and bottom arrows are the input, and the right and top arrows are the output. Then

$$(R_{ij})_{\rightarrow\downarrow, \leftarrow\downarrow} = \begin{cases} w \left(\begin{array}{c} \text{?} \\ \text{?} \end{array} \right) & \text{if number of incoming = outgoing,} \\ 0 & \text{otherwise.} \end{cases}$$

Start with one periodic strip.



Define

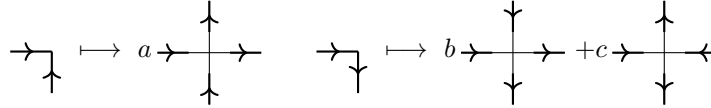
$$R_{0j} : V_0 \otimes V_j \rightarrow V_0 \otimes V_j$$

via the matrix

	$\rightarrow\uparrow$	$\rightarrow\downarrow$	$\leftarrow\uparrow$	$\leftarrow\downarrow$
$\rightarrow\uparrow$	a	0	0	0
$\rightarrow\downarrow$	0	b	c	0
$\leftarrow\uparrow$	0	c	b	0
$\leftarrow\downarrow$	0	0	0	a

For example,

$$\begin{aligned} R_{0j}(\rightarrow\uparrow) &= a(\rightarrow\uparrow), \\ R_{0j}(\rightarrow\downarrow) &= b(\rightarrow\downarrow) + c(\leftarrow\uparrow). \end{aligned}$$



Now we define a $2^{\ell+1} \times 2^{\ell+1}$ matrix

$$R_{0\ell} \cdots R_{02} R_{01} : V_0 \otimes V_1 \otimes \cdots \otimes V_\ell \rightarrow V_0 \otimes V_1 \otimes \cdots \otimes V_\ell$$

where R_{0j} acts on factors $0, j$ and as the identity on all other factors. Thus this is the product

$$(R_{0\ell} \otimes \text{Id}_{[\ell] \setminus \{\ell\}}) \cdots (R_{02} \otimes \text{Id}_{[\ell] \setminus \{2\}})(R_{01} \otimes \text{Id}_{[\ell] \setminus \{1\}}),$$

with the tensor factors placed in their original order.

For example,

Since we're considering the torus, we want only matching horizontal arrows at the two ends:

$$\rightarrow \text{---} \rightarrow \quad \text{or} \quad \leftarrow \text{---} \leftarrow$$

Definition 2.1. Let X be an operator on $V_0 \otimes W$. Write its blocks using the input arrow first and the output arrow second:

$$X = \begin{pmatrix} X_{\rightarrow\rightarrow} & X_{\leftarrow\rightarrow} \\ X_{\rightarrow\leftarrow} & X_{\leftarrow\leftarrow} \end{pmatrix}.$$

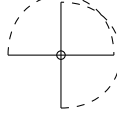
The **partial trace** of X is

$$\text{tr}_0 X = X_{\rightarrow\rightarrow} + X_{\leftarrow\leftarrow}.$$

The partial trace is the unique linear map such that $\text{tr}_0(X_0 \otimes Y) = \text{tr}(X_0)Y$.

The **transfer matrix** for a periodic strip of length ℓ is $T_\ell = \text{tr}_0(R_{0\ell} \cdots R_{02}R_{01})$. Taking the m th power, the matrix T_ℓ^m corresponds to stacking m rows. Taking the trace $\text{tr}(T_\ell^m)$ then corresponds to identifying the top and bottom boundaries.

Example 2.2 ($m = \ell = 1$).



If $m = \ell = 1$, then

$$\begin{aligned} T_1 &= \text{tr}_0(R_{01}) \\ &= \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} + \begin{pmatrix} b & 0 \\ 0 & a \end{pmatrix} \\ &= \begin{pmatrix} a+b & 0 \\ 0 & a+b \end{pmatrix}. \end{aligned}$$

In this case,

$$Z = \text{tr}(T_1) = 2(a+b).$$

This is also equal to

$$w(\begin{array}{c} \uparrow \\ \rightarrow \\ \downarrow \\ \leftarrow \end{array}) + w(\begin{array}{c} \uparrow \\ \leftarrow \\ \downarrow \\ \rightarrow \end{array}) + w(\begin{array}{c} \leftarrow \\ \uparrow \\ \leftarrow \\ \downarrow \end{array}) + w(\begin{array}{c} \leftarrow \\ \downarrow \\ \leftarrow \\ \uparrow \end{array}).$$

3 Properties & Integrability

Now we will reparametrize a, b, c in terms of variables $\{u_i\}$ and q , where each line i will contribute a variable u_i . The variable q is called the **quantum parameter**.

Each vertex witnesses some crossing of line i by line j . To this vertex, we attach the **spectral parameter** u_j/u_i . We get weights that look like

$$\begin{cases} a_{ij} = \frac{qu_j}{u_i} - \frac{u_i}{qu_j}, \\ b_{ij} = \frac{u_j}{u_i} - \frac{u_i}{u_j}, \\ c_{ij} = q - q^{-1}. \end{cases}$$

If u_j/u_i is constant at all crossings, then this is called the **homogeneous** six-vertex model.

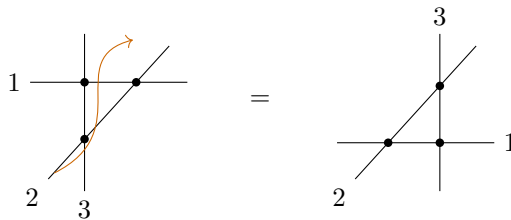
We call the combination $\Delta = \frac{a^2+b^2-c^2}{2ab} = \frac{q+q^{-1}}{2}$ the **anisotropy parameter**. The combinatorial interpretations of Z discussed earlier are each valid for a certain specialization of Δ to a real number. For ASMs and 3-colorings, $\Delta = \frac{1}{2}$. For domino tilings, $\Delta = 0$.

There are two important identities satisfied by the R -matrices.

Yang-Baxter Equation

We write R_{ij} as shorthand for $R_{ij}(\frac{u_j}{u_i})$. The **Yang-Baxter equation** is the identity

$$R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}.$$



Unitarity Equation

$$R_{12}(x)R_{21}(x^{-1}) = (qx - q^{-1}x^{-1})(qx^{-1} - q^{-1}x)\text{Id}, \quad x = u_2/u_1.$$

$$\begin{array}{c} 2 \\ \diagdown \quad \diagup \\ 1 \end{array} \quad \propto \quad \begin{array}{c} 2 \\ \text{---} \\ 1 \end{array}$$

Now attach parameters x_i to the i th horizontal line, and y_j to the j th vertical line. Then by the Yang-Baxter equation,

$$[T_\ell(x_i; y_1, \dots, y_\ell), T_\ell(x_{i'}; y_1, \dots, y_\ell)] = 0.$$

This commuting family of transfer matrices expresses the integrability of the six-vertex model.

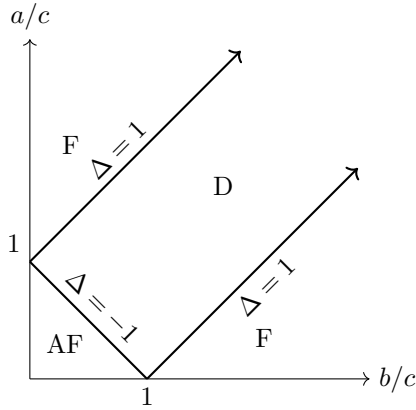
In pictures, for $m = \ell = 2$:

$$\begin{array}{ccccc} \begin{array}{c} y_1 \ y_2 \\ | \quad | \\ x_1 \text{---} \text{---} \\ | \quad | \\ x_2 \text{---} \text{---} \end{array} & \equiv \text{U} & \begin{array}{c} y_1 \ y_2 \\ | \quad | \\ x_1 \text{---} \text{---} \\ | \quad | \\ x_2 \text{---} \text{---} \end{array} & \equiv \text{YB} & \begin{array}{c} y_1 \ y_2 \\ | \quad | \\ x_1 \text{---} \text{---} \\ | \quad | \\ x_2 \text{---} \text{---} \end{array} \\ & & & & \\ \begin{array}{c} y_1 \ y_2 \\ | \quad | \\ x_1 \text{---} \text{---} \\ | \quad | \\ x_2 \text{---} \text{---} \end{array} & \equiv \text{YB} & \begin{array}{c} y_1 \ y_2 \\ | \quad | \\ x_1 \text{---} \text{---} \\ | \quad | \\ x_2 \text{---} \text{---} \end{array} & \equiv \text{U} & \begin{array}{c} y_1 \ y_2 \\ | \quad | \\ x_2 \text{---} \text{---} \\ | \quad | \\ x_1 \text{---} \text{---} \end{array} \end{array}$$

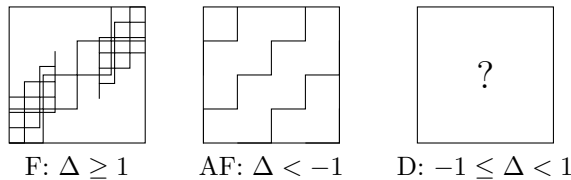
Here U means inserting or removing two crossings using unitarity (with its scalar factor), and YB means moving a crossing past a vertical line. The horizontal lines are periodic.

4 The phase diagram

In this section, we focus on the homogeneous six-vertex model with positive weights. We can draw a phase diagram in parameter space, which has three interesting **phases**: the ferroelectric phase, the antiferroelectric phase, and the disordered phase.



The behavior of a probable state in each of these phases, viewed in the path model, looks schematically like:



The ferroelectric phase is frozen, the antiferroelectric phase has alternating local order, and the disordered phase has fluctuations at all scales.